PASSIVE COOLING SYSTEM FOR HOME

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Project Abstract

This project presents a Passive Cooling System for Home designed to reduce indoor temperature without relying on conventional air-conditioning systems. The system utilizes natural cooling techniques such as natural ventilation, shading, thermal insulation, and reflective surfaces to minimize heat gain and improve indoor thermal comfort. An ESP32-based monitoring system with a DS18B20 temperature sensor, RTC module, and SD card is used to monitor and record temperature data. The system provides an eco-friendly, energy-efficient, and cost-effective approach for residential cooling. The project achieved an indoor temperature reduction of approximately 2°C under the tested conditions.

Problem Statement

Conventional cooling systems consume high electrical energy and increase electricity costs while contributing to environmental impact . There is a need for a sustainable passive cooling solution that reduces indoor temperature and improves thermal comfort without mechanical cooling.

Need for the Project

The increasing use of conventional cooling systems leads to higher energy consumption and electricity costs, especially during hot weather. Passive cooling provides a sustainable and cost-effective solution by reducing indoor temperature through natural ventilation, shading, insulation, and reflective surfaces.

Project Objectives

- Reduce indoor temperature using passive cooling techniques.
- Improve indoor thermal comfort and reduce heat gain.
- Reduce energy consumption and dependence on mechanical cooling.

Expected Learning Outcomes

- Understand passive cooling techniques and thermal comfort.
- Gain practical skills in temperature monitoring and data logging.

System Requirements

- **Hardware:** ESP32, DS18B20 temperature sensor, DS3231 RTC, SD card module, SD card, jumper wires, and 4.7 kΩ pull-up resistor.
- **Software:** ESP32 programming environment for temperature monitoring and data logging.

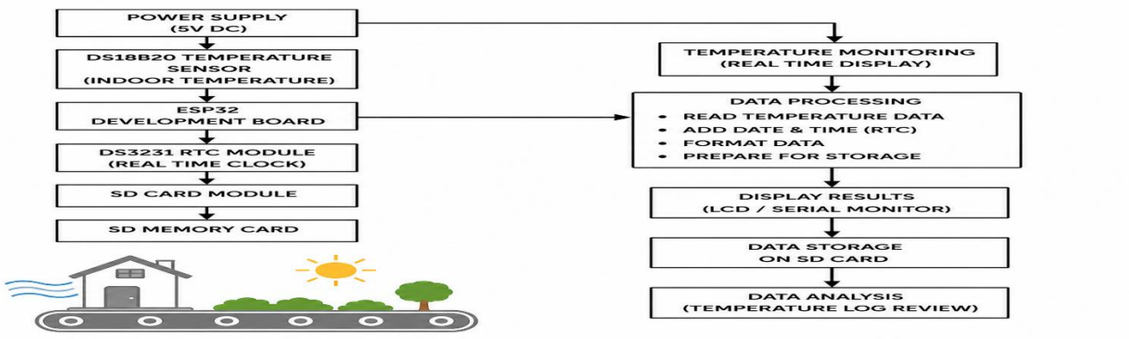
Methodology / Workflow

Climatic Data Collection → Passive Cooling Design → Ventilation & Shading → Insulation & Reflective Surfaces → Temperature Monitoring → Data Logging → Performance Evaluation.

System Architecture

The DS18B20 sensor measures indoor temperature, while the ESP32 processes the data and the DS3231 RTC provides time stamps. The processed temperature data is monitored in real time and stored in an SD card for analysis.

BLOCK DIAGRAM



Project Development Process

The project was developed by identifying the cooling problem, studying passive cooling techniques, and designing an integrated system using ventilation, shading, insulation, and reflective surfaces. The prototype was then implemented with temperature monitoring and data logging, followed by testing and performance evaluation.

Implementation

The passive cooling techniques were implemented by integrating natural ventilation, shading, thermal insulation, and reflective surfaces into the residential setup. The temperature monitoring system using ESP32, DS18B20, DS3231 RTC, and SD card was then used to record and evaluate the cooling performance.

Algorithms / Models Used

- **Passive Cooling Techniques** – Used for natural ventilation, shading, thermal insulation, and reflective heat reduction.
- **Temperature Monitoring & Data Logging** – Used for continuous indoor temperature measurement and performance analysis using ESP32 and DS18B20.

[Insert QR Code]

QR Code — Project Report (PDF) | Source Code Repository | Demonstration Video

Experimental Setup

The prototype consists of ESP32, DS18B20 temperature sensor, DS3231 RTC module, and SD card module for continuous temperature monitoring and data logging. The system was tested under summer conditions with an ambient temperature of 35°C, and temperature readings were recorded at 1-minute intervals.

Results

The passive cooling system achieved an approximate 2°C reduction in indoor temperature.
The system improved thermal comfort, reduced heat gain, and minimized dependence on mechanical cooling.

Performance Evaluation

The system performance was evaluated by comparing indoor temperature before and after implementing passive cooling techniques. The system achieved an approximate 2°C reduction in indoor temperature, demonstrating improved thermal comfort and reduced dependence on mechanical cooling.

Testing and Validation

The system was tested by continuously measuring indoor temperature using the DS18B20 sensor and recording the readings with date and time. The results were compared before and after implementing passive cooling techniques to validate the temperature reduction and cooling performance.

Challenges Encountered

- Variation in climatic conditions affected cooling performance.
- Natural airflow and solar heat gain were difficult to control.

Solutions Implemented

- **Passive cooling techniques** – Used ventilation, shading, insulation, and reflective surfaces to reduce heat gain.
- **Temperature monitoring** – Used ESP32 and DS18B20 to continuously monitor and record indoor temperature.

Innovation / Novel Features

The project integrates passive cooling techniques with real-time temperature monitoring and data logging, providing a simple, low-power, and sustainable cooling solution for homes.

Applications

- Residential homes in hot climatic regions.
- Energy-efficient and sustainable building.

Future Scope

- Integrate IoT and mobile monitoring for real-time temperature tracking.
- Use Phase Change Materials (PCM) to improve cooling efficiency..
- Develop automatic ventilation control using smart sensors.
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Project Outcomes

The system achieved an approximate 2°C reduction in indoor temperature and improved thermal comfort. It reduced dependence on conventional cooling and supported energy conservation. The project demonstrated a simple, cost-effective, and eco-friendly cooling solution.

Skills Acquired

- **Technical Skill:** Passive Cooling System Design, ESP 32 & Sensor Interfacing, Temperature Monitoring & Data Logging.
- **Soft Skill:** Problem Solving, Team Work, Project Planning

References

- ***Givoni,*** *B., Passive and Low Energy Cooling of Buildings, John Wiley & Sons, 1994..*
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